Supplementary Material for:

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the Brown Note

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1 Detailed Derivations

This section provides the full derivation of all analytical results used in the main text. Equation numbering follows the convention (Sn) to distinguish supplementary equations from those in the main paper.

1.1 Shell Flexural Rigidity

The abdominal wall is modelled as a thin isotropic shell of thickness h, Young's modulus E, and Poisson's ratio ν. From classical Love–Kirchhoff thin-shell theory, the flexural (bending) rigidity is

$$D = \frac{E\,h^3}{12\,(1-\nu^2)}\,. \tag{S1}$$

For the canonical parameter set ($E = 0.1\,\mathrm{MPa}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$):

$$D = \frac{0.1\times 10^6\ \times\ (0.01)^3}{12\,(1-0.45^2)} = \frac{10^5\times 10^{-6}}{12\times 0.7975} \approx 0.0104\,\mathrm{N\,m}\,.$$

The thin-shell condition $h/R \approx 0.06 \ll 1$ is satisfied for the equivalent-sphere radius $R_{\mathrm{eq}} = (a^2c)^{1/3} \approx 0.157\,\mathrm{m}$.

1.2 Added Mass Coefficient for Mode n (Lamb's Result)

For flexural mode n ($n \geq 2$) of a thin spherical shell of radius R filled with an incompressible fluid of density ρ_f, the interior velocity potential is expanded in Legendre polynomials. The solution to Laplace's equation inside a sphere for mode n gives a radial dependence $\phi \propto r^n$, so the fluid velocity at the shell wall scales as r^{n-1}.

Integrating the kinetic energy of the internal flow, Lamb (1882) showed that the fluid added mass per unit shell area is

$$m_{\mathrm{add}}^{(n)} = \frac{\rho_f\,R}{n}\,. \tag{S2}$$

This result has a transparent physical interpretation: for higher mode numbers n, the fluid motion is more localised near the wall (shorter "penetration depth" $\sim R/n$), entraining less fluid mass.

For the $n = 2$ oblate–prolate mode with $\rho_f = 1020\,\mathrm{kg/m^3}$ and $R = 0.157\,\mathrm{m}$:

$$m_{\mathrm{add}}^{(2)} = \frac{1020\times 0.157}{2} = 80.1\,\mathrm{kg/m^2}\,.$$

Compare this with the shell wall mass per unit area, $m_{\mathrm{wall}} = \rho_w\,h = 1100\times 0.01 = 11.0\,\mathrm{kg/m^2}$. The fluid added mass exceeds the wall mass by a factor of $\sim$7.3, confirming that fluid inertia dominates the effective modal mass.

1.3 Flexural Frequency Formula with Intermediate Steps

The flexural modes ($n \geq 2$) involve shell deformation at approximately constant enclosed volume; the fluid acts as added mass but does not contribute volumetric (bulk) stiffness. The restoring force per unit shell area comprises three terms:

Bending stiffness. From Love–Kirchhoff theory for a spherical shell:

$$K_{\text{bend}}^{(n)} = \frac{n\,(n-1)\,(n+2)^2\,D}{R^4}\,. \tag{S3}$$

Membrane stiffness. Lamb (1882) obtained the membrane contribution:

$$K_{\text{memb}}^{(n)} = \frac{E\,h}{R^2}\,\lambda_n\,, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}\,. \tag{S4}$$

Pre-stress stiffness. The intra-abdominal pressure P_{iap} prestresses the shell in biaxial tension, adding a geometric stiffness:

$$K_P^{(n)} = \frac{P_{\text{iap}}}{R}\,(n-1)(n+2)\,. \tag{S5}$$

Total stiffness and effective mass. The total stiffness per unit area is

$$K_{\text{total}}^{(n)} = K_{\text{bend}}^{(n)} + K_{\text{memb}}^{(n)} + K_P^{(n)}\,, \tag{S6}$$

and the effective mass per unit area is

$$m_{\text{eff}}^{(n)} = \rho_w\,h + \frac{\rho_f\,R}{n}\,. \tag{S7}$$

The natural frequency follows as

$$\boxed{f_n = \frac{1}{2\pi}\sqrt{\frac{K_{\text{total}}^{(n)}}{m_{\text{eff}}^{(n)}}}}\,. \tag{S8}$$

Numerical evaluation for $n = 2$. With the canonical parameters ($E = 0.1\,\text{MPa}$, $a = 0.18\,\text{m}$, $c/a = 0.67$, $h = 10\,\text{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\text{kg/m}^3$, $\rho_f = 1020\,\text{kg/m}^3$, $P_{\text{iap}} =$

1000 Pa):

$$
\begin{aligned}
R_{\text{eq}} &= (0.18^2 \times 0.12)^{1/3} = 0.157\,\text{m}, \\
D &= 0.0104\,\text{N}\,\text{m}, \\
K^{(2)}_{\text{bend}} &= \frac{2 \times 1 \times 16 \times 0.0104}{0.157^4} = 547\,\text{Pa/m}, \\
\lambda_2 &= \frac{4+2-2+0.90}{4+2+1-0.45} = \frac{4.90}{6.55} = 0.748, \\
K^{(2)}_{\text{memb}} &= \frac{10^5 \times 0.01}{0.157^2} \times 0.748 = 30\,363\,\text{Pa/m}, \\
K^{(2)}_{P} &= \frac{1000}{0.157} \times 1 \times 4 = 25\,478\,\text{Pa/m}, \\
K^{(2)}_{\text{total}} &= 547 + 30\,363 + 25\,478 = 56\,388\,\text{Pa/m}, \\
m^{(2)}_{\text{eff}} &= 1100 \times 0.01 + \frac{1020 \times 0.157}{2} = 11.0 + 80.1 = 91.1\,\text{kg/m}^2, \\
f_2 &= \frac{1}{2\pi}\sqrt{\frac{56\,388}{91.1}} \approx 4.0\,\text{Hz}\,.
\end{aligned}
$$

1.4 Breathing Mode Derivation

The breathing mode ($n = 0$) involves a uniform radial expansion/contraction of the shell. Unlike flexural modes, this motion *does* compress the enclosed fluid, so the fluid bulk modulus K_f contributes directly to the stiffness.

Shell membrane stiffness. For a uniform radial displacement δR, the circumferential strain is $\varepsilon = \delta R/R$. Biaxial membrane stress gives a restoring pressure:

$$k_{\text{shell}} = \frac{2\,E\,h}{R^2\,(1-\nu)}\,. \tag{S9}$$

Fluid volumetric stiffness. The volume change is $\Delta V = 4\pi R^2\,\delta R$, and the pressure change is $\Delta p = K_f\,\Delta V/V = 3\,K_f\,\delta R/R$. The restoring force per unit shell area is therefore

$$k_{\text{fluid}} = \frac{3\,K_f}{R}\,. \tag{S10}$$

Comparison of stiffness contributions.

$$
\begin{aligned}
k_{\text{shell}} &= \frac{2 \times 10^5 \times 0.01}{0.157^2 \times 0.55} \approx 2.3 \times 10^5\,\text{Pa/m}, \\
k_{\text{fluid}} &= \frac{3 \times 2.2 \times 10^9}{0.157} \approx 4.2 \times 10^{10}\,\text{Pa/m}.
\end{aligned}
$$

The fluid volumetric stiffness exceeds the shell membrane stiffness by a factor of $\sim 1.8 \times 10^5$; hence K_f completely dominates the breathing mode stiffness.

Effective mass and frequency.

$$m_{\text{eff}}^{(0)} = \rho_w\, h + \rho_f\, R = 11.0 + 160.1 = 171.1\,\text{kg/m}^2, \tag{S11}$$

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{m_{\text{eff}}^{(0)}}} \approx \frac{1}{2\pi}\sqrt{\frac{4.2\times 10^{10}}{171.1}} \approx 2900\,\text{Hz}\,. \tag{S12}$$

This is far above the infrasonic range ($< 20\,\text{Hz}$) and is therefore irrelevant to the brown-note hypothesis.

1.5 Energy-Consistent Displacement Derivation

The pressure-based estimate of airborne coupling displacement (main text, Section 4) overestimates the true value by ∼40% because it does not enforce energy conservation. The self-consistent derivation, following the reciprocity formulation of Junger & Feit (2012), proceeds as follows.

Step 1: Absorption cross-section. By reciprocity, the maximum absorption cross-section at resonance for a mode of order n is

$$\sigma_{\text{max}}^{(n)} = \frac{(2n+1)\,\lambda^2}{4\pi}\,, \tag{S13}$$

where $\lambda = c_{\text{air}}/f_n$ is the acoustic wavelength. The actual cross-section is limited by the ratio of radiation damping to total damping:

$$\sigma_{\text{abs}}^{(n)} = \sigma_{\text{max}}^{(n)} \times \frac{\zeta_{\text{rad}}}{\zeta_{\text{rad}} + \zeta_{\text{struct}}}\,. \tag{S14}$$

Step 2: Radiation damping. The radiation resistance for mode n of a sphere in the Rayleigh limit ($ka \ll 1$) is

$$R_{\text{rad}}^{(n)} = \rho_{\text{air}}\, c_{\text{air}}\, 4\pi R^2 \times \frac{(ka)^{2n+2}}{[(2n+1)!!]^2}\,, \tag{S15}$$

where $(2n+1)!! = 1\times 3\times 5\times\cdots\times(2n+1)$ is the double factorial. For $n = 2$ at $f = 4\,\text{Hz}$:

$$\begin{gathered} ka = \frac{2\pi\times 4\times 0.157}{343} \approx 0.012, \\ (ka)^6 \approx 2.4\times 10^{-12}, \qquad 5!! = 15, \\ \zeta_{\text{rad}} \sim 10^{-15} \ll \zeta_{\text{struct}} = 0.125\,. \end{gathered}$$

Step 3: Steady-state energy balance. At steady state, absorbed power equals dissipated power:

$$\sigma_{\text{abs}}\, I = \zeta_{\text{struct}}\, \omega_n^3\, M_{\text{total}}\, \xi^2\,, \tag{S16}$$

where $I = p_{\text{inc}}^2/(2\rho_{\text{air}} c_{\text{air}})$ is the incident intensity, $M_{\text{total}} = m_{\text{eff}} \times 4\pi R^2$ is the total modal mass, and ξ is the displacement amplitude.

Solving for the energy-consistent displacement:

$$\boxed{\xi_{\text{energy}} = \sqrt{\frac{\sigma_{\text{abs}}\, I}{\zeta_{\text{struct}}\, \omega_n^3\, M_{\text{total}}}}}. \tag{S17}$$

Comparison with the pressure-based estimate. At 120 dB SPL and $n = 2$:

$$\xi_{\text{energy}} \approx 0.014\,\mu\text{m},$$
$$\xi_{\text{pressure}} \approx 0.18\,\mu\text{m}.$$

The pressure-based estimate is approximately 12.9× larger because it does not account for the extremely low radiation efficiency ($\zeta_{\text{rad}}/\zeta_{\text{struct}} \sim 10^{-14}$). Both values are far below the PIEZO1/2 mechanotransduction threshold (0.5 µm to 2.0 µm), so the overestimate does not affect any conclusions.

2 Complete Parameter Tables

2.1 Canonical Parameter Set

Table 1 lists the baseline parameter values used throughout the analysis, together with literature sources.

Table 1: Canonical (baseline) parameter set.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Young's modulus	E	0.1	MPa	Relaxed passive tissue; Fung (1993)
Poisson's ratio	ν	0.45	—	Nearly incompressible
Wall density	ρ_w	1100	kg/m^3	Soft tissue
Fluid density	ρ_f	1020	kg/m^3	Abdominal fluid
Fluid bulk modulus	K_f	2.2	GPa	Water at 37 °C
Intra-abdominal pressure	P_{iap}	1000	Pa	Resting supine; ∼7.5 mmHg
Loss tangent	η	0.25	—	Viscoelastic tissue
Derived quantities				
Equivalent sphere radius	R_{eq}	0.157	m	$(a^2c)^{1/3}$
Flexural rigidity	D	0.0104	N m	Eq. (S1)
Damping ratio	ζ	0.125	—	$\eta/2$
Quality factor	Q	4.0	—	$1/(2\zeta)$

2.2 Monte Carlo Parameter Distributions

Table 2 specifies the probability distributions used in the uncertainty quantification (N = 10,000 Monte Carlo samples, Sobol base $N_{\text{Sobol}} = 4096$, seed = 42).

Table 2: Parameter distributions for Monte Carlo / Sobol analysis.

Parameter	Nominal	Distribution	Low	High
E (Pa)	10^5	Lognormal	5×10^4	2×10^6
a (m)	0.18	Normal	0.15	0.22
c (m)	0.12	Normal	0.08	0.15
h (m)	0.01	Normal	0.005	0.02
ν	0.45	Uniform	0.40	0.499
ρ_w (kg/m^3)	1100	Normal	1000	1200
ρ_f (kg/m^3)	1020	Normal	1000	1060
P_{iap} (Pa)	1000	Lognormal	500	3000
η	0.25	Uniform	0.10	0.40

2.3 Complete Sobol Sensitivity Indices

Tables 3–6 present the first-order (S_1) and total-order (S_T) Sobol indices for each output quantity.

Table 3: Sobol indices for $n = 2$ flexural frequency f_2.

Parameter	S_1	S_T
E	0.8119	0.8613
a	0.0353	0.0503
c	0.0272	0.0359
h	0.0659	0.1163
ν	0.000 13	0.000 24
ρ_w	0.000 079	0.000 093
ρ_f	0.000 033	0.000 24
P_{iap}	0.0108	0.0125
η	0.0000	0.0000

2.4 Elastic Modulus Sweep

Table 7 shows the sensitivity of the fundamental flexural frequency to the elastic modulus, the parameter that dominates modal uncertainty ($S_T = 0.86$).

Table 4: Sobol first-order indices for airborne displacement ξ_{air} at 120 dB.

Parameter	S_1
E	< 0.001
a	0.041
c	0.028
h	0.011
ν	< 0.001
ρ_w	< 0.001
ρ_f	0.001
P_{iap}	< 0.001
η	0.899

Table 5: Sobol indices for mechanical displacement ξ_{mech} at $0.5\,\text{m/s}^2$.

Parameter	S_1
E	0.244
a	0.015
c	0.014
h	0.102
ν	< 0.001
ρ_w	< 0.001
ρ_f	< 0.001
P_{iap}	0.022
η	0.089

Table 6: Sobol indices for coupling ratio $\mathcal{R} = \xi_{\text{mech}}/\xi_{\text{air}}$.

Parameter	S_1
E	0.331
a	0.005
c	0.006
h	0.088
ν	< 0.001
ρ_w	< 0.001
ρ_f	0.001
P_{iap}	0.027
η	< 0.001

Table 7: $n = 2$ frequency vs. elastic modulus.

E (MPa)	f_2 (Hz)	ISO 2631 (4–8 Hz)?
0.05	3.4	Marginal
0.10	4.0	Yes
0.20	4.9	Yes
0.50	7.1	Yes
1.00	9.6	Marginal
2.00	13.3	No

2.5 Parametric Study Overview

A full factorial parametric study explored 486 combinations: 6 values of E × 3 semi-major axes × 3 aspect ratios × 3 wall thicknesses × 3 loss tangents.

Key findings from the 486-combination sweep:

- 37% of configurations place f_2 in the 5–10 Hz range.
- 41% of configurations place f_2 in the ISO 2631 range (4–8 Hz).
- Dominant factor: E, followed by geometry (a, c/a) and P_{iap}.

Table 8: Representative body-type configurations and ISO 2631 overlap.

Body type	E (MPa)	a (cm)	f_2 (Hz)	In ISO range?
Lean	0.2	15	6.3	Yes
Average	0.1	18	4.0	Yes
Large	0.08	20	3.1	Marginal

2.6 Monte Carlo Summary Statistics

Table 9 summarises the full Monte Carlo output distributions ($N = 10{,}000$ samples).

Table 9: Monte Carlo summary statistics ($N = 10{,}000$).

Quantity	Mean	Median	Std Dev	P5	P95	99% CI upper
f_2 (Hz)	7.52	6.43	4.21	3.32	15.48	26.43
ξ_{air} (µm)	0.206	0.178	0.090	0.107	0.394	0.490
ξ_{mech} (µm)	2745	1818	3110	296	8176	17 457
$\mathcal{R}$	12 949	9820	11 888	1780	34 169	62 466
$\text{SPL}_{\text{PIEZO}}$ (dB)	134.4	135.0	3.5	128.1	139.4	141.1

3 Sensitivity to Modelling Assumptions

This section summarises validation studies that quantify the error introduced by each simplifying assumption in the analytical model.

3.1 Oblate Spheroid vs. Equivalent Sphere

The main-text model uses an equivalent sphere of radius $R_{\text{eq}} = (a^2c)^{1/3}$. A Rayleigh–Ritz analysis on the full oblate spheroid (Section 1) was performed using `oblate_spheroid_ritz.py` with a 2-DOF trial function per mode (normal and tangential displacements expressed as Legendre polynomials), integrated over the oblate surface via 200-point Gauss–Legendre quadrature.

Table 10: Sphere vs. oblate Rayleigh–Ritz frequencies at $c/a = 0.67$.

	$E = 0.1\,\text{MPa}$		$E = 0.5\,\text{MPa}$	
Mode	Sphere (Hz)	Ritz (Hz)	Sphere (Hz)	Ritz (Hz)
$n = 2$	4.0	3.6	7.1	6.0
$n = 3$	6.3	5.6	11.3	9.5
$n = 4$	8.9	8.0	14.4	12.9

Findings. The sphere model overestimates flexural frequencies by 11–20%. The error grows with eccentricity and with n. Physical reasons: (i) spatially varying curvature on the oblate surface (poles sharper, equator flatter), (ii) different fluid added-mass distribution from oblate flow geometry, and (iii) the Ritz formulation captures optimal tangential displacement via the variational principle. Despite the systematic offset, the conclusion that f_2 lies in the 3 Hz to 8 Hz range is robust to the choice of geometry model.

3.2 Multilayer vs. Homogeneous Wall

The real abdominal wall comprises ∼5–6 distinct layers. Table 11 shows the layer stack used in `multilayer_wall.py`.

Effective composite properties are computed using classical laminate theory (Hernández-Gascón et al., 2013):

$$(Eh)_{\text{eff}} = \sum_i E_i\, h_i\,, \tag{S18}$$

$$D_{\text{eff}} = \sum_i \frac{E_i}{1-\nu_i^2}\left[\frac{h_i^3}{12} + h_i(z_i - \bar{z})^2\right], \tag{S19}$$

$$\rho_{\text{eff}} = \frac{\sum_i \rho_i\, h_i}{\sum_i h_i}\,. \tag{S20}$$

Table 11: Relaxed abdominal wall layer properties.

Layer	h (mm)	E (MPa)	ν	ρ (kg/m^3)
Peritoneum	0.1	0.50	0.45	1050
Transversalis	3.0	0.05	0.45	1050
Internal oblique	5.0	0.05	0.45	1060
External oblique	5.0	0.08	0.45	1060
Subcutaneous fat	10.0	0.002	0.49	950
Skin	2.0	0.50	0.45	1100
Composite	**25.1**	**0.075**	**0.473**	**1012**

Key result. The multilayer relaxed model predicts $f_2 \approx 4.6\,\mathrm{Hz}$, within 4% of the homogeneous model at $E = 0.1\,\mathrm{MPa}$. The tensed muscle configuration ($E_{\mathrm{eff}} \approx 2.7\,\mathrm{MPa}$) raises f_2 to $\sim$23 Hz, confirming that the dominant variable is muscle tone, not the specific layer arrangement.

3.3 Viscous vs. Inviscid Fluid

Fluid viscosity introduces a Stokes oscillatory boundary layer of thickness

$$\delta = \sqrt{\frac{2\mu}{\rho_f\,\omega}} \tag{S21}$$

at the inner wall. The associated viscous damping ratio addition is (Landau & Lifshitz, §24)

$$\Delta\zeta_{\mathrm{visc}} = \frac{\rho_f\,\delta\,(n+1)}{4\,n\,(2n+1)\,m_{\mathrm{eff}}^{(n)}}\,. \tag{S22}$$

Table 12: Viscous correction for physiological fluids ($n = 2$, $f = 4\,\mathrm{Hz}$).

Fluid	μ (Pa s)	δ (mm)	$\Delta\zeta_{\mathrm{visc}}$	$\Delta\zeta/\zeta_{\mathrm{struct}}$ (%)	ΔQ (%)
Water	0.001	$\sim$0.2	$\sim 10^{-4}$	< 0.1	< 0.1
Bile	0.005	$\sim$0.4	$\sim 3\times10^{-4}$	< 0.3	< 0.3
Blood	0.004	$\sim$0.4	$\sim 2\times10^{-4}$	< 0.2	< 0.2
Gastric acid	0.01	$\sim$0.6	$\sim 4\times10^{-4}$	< 0.4	< 0.4
Mucus	0.10	$\sim$2.0	$\sim 10^{-3}$	< 1.0	< 1.0

Key result. Even for the most viscous physiological fluid (mucus, $\mu = 0.1\,\mathrm{Pa\,s}$), the viscous damping adds $< 1\%$ to the structural damping. The crossover viscosity at which $\Delta\zeta_{\mathrm{visc}} = \zeta_{\mathrm{struct}}$ is $\mu_{\mathrm{cross}} \approx 280\,\mathrm{Pa\,s}$—comparable to honey or peanut butter and far beyond any abdominal fluid. The inviscid approximation is therefore well justified.

Non-Newtonian effects. Chyme is shear-thinning (power-law exponent $n_{\rm pl} \approx 0.5$–0.8). At the extremely low shear rates characteristic of infrasonic oscillation ($\dot{\gamma} \sim 0.1/\mathrm{s}$), the apparent viscosity is *higher* than the consistency index, but the viscous damping correction remains negligible ($\Delta\zeta/\zeta_{\rm struct} < 2\%$).

3.4 With and Without Organ Inclusions

Solid organs (liver, spleen, kidneys) occupy ∼20–30% of the cavity volume with shear moduli of 1–10 kPa.

The analysis (`organ_inclusions.py`) uses Hashin–Shtrikman bounds for the effective medium. Key results:

1. **Bulk modulus**: Since both fluid and tissue have $K \approx 2.2\,\mathrm{GPa}$ (nearly incompressible), the effective bulk modulus changes by $< 1\,\mathrm{ppm}$—negligible.
2. **Shear modulus**: The HS lower bound gives $G_{\rm eff} = 0$ for the composite because the fluid matrix cannot transmit shear at the macroscale (inclusions are non-percolating, $\phi < 0.29$).
3. **Density**: The effective density is a simple volume average; organ tissue ($\rho \approx 1055\,\mathrm{kg/m^3}$) vs. fluid ($\rho = 1020\,\mathrm{kg/m^3}$) gives $\Delta\rho/\rho \approx 1\%$.
4. **Frequency shift**: The combined stiffness and mass perturbation shifts f_2 by <1%. The liver's asymmetric mass (1.5 kg) lifts the m-degeneracy of the $n = 2$ mode by ∼1–2%.

Conclusion. Treating the cavity contents as homogeneous fluid introduces $< 2\%$ frequency error, well within the uncertainty from the elastic modulus ($\sigma_E/E \sim$ factor of 5).

3.5 Nonlinear vs. Linear Analysis

For whole-body vibration at occupational levels, the predicted displacements ($\xi \sim 650$–$7500\,\mu\mathrm{m}$) approach $w/h \sim O(1)$, suggesting geometric nonlinearity may be relevant. The analysis (`nonlinear_analysis.py`) constructs a Duffing oscillator model from Donnell–von Kármán nonlinear shell theory.

Cubic stiffness coefficient. The von Kármán mid-surface strains introduce a cubic restoring force with coefficient

$$\alpha = \alpha_{\rm shell} + \alpha_{\rm fluid}\,, \tag{S23}$$

where

$$\alpha_{\rm shell} = \frac{E\,h}{4\,R^4}\,n^2(n+1)^2\,C_{\rm NL}\,, \quad C_{\rm NL} = \frac{3-\nu^2}{1-\nu^2}\,, \tag{S24}$$

$$\alpha_{\rm fluid} = \frac{3\,\rho_f\,\omega_0^2}{2\,R^2\,n}\,. \tag{S25}$$

The system is *hardening* ($\alpha > 0$): frequency increases with amplitude. For the $n = 2$ mode, $\alpha \approx 7.1 \times 10^7\,\mathrm{rad^2 s^{-2} m^{-2}}$.

Backbone curve. The amplitude-dependent resonance frequency is

$$\omega(A) = \omega_0 \sqrt{1 + \frac{3\,\alpha\,A^2}{4\,\omega_0^2}}\,. \tag{S26}$$

A 5% frequency shift occurs at a critical amplitude of $A_{\text{crit}} \approx 1092\,\mu\text{m}$ ($A/h \approx 0.11$).

Amplitude-dependent damping. Soft tissue exhibits increased dissipation at large strains, modelled as

$$\eta(A) = \eta_0\big(1 + \beta|A/h|^2\big)\,, \quad \beta = 0.5\,. \tag{S27}$$

This partially offsets the nonlinear stiffening by reducing peak amplitudes.

Key result. For airborne coupling ($\xi < 0.2\,\mu\text{m}$), nonlinearity is completely negligible ($A/h \sim 10^{-5}$). For mechanical coupling at occupational WBV levels, the nonlinear frequency shift is 5–15%, which stiffens the response and slightly reduces peak displacement but does not change the order-of-magnitude conclusion ($\xi_{\text{mech}} \gg$ PIEZO threshold).

3.6 Boundary Condition Effects

The Rayleigh–Ritz framework also investigated boundary condition sensitivity via FEA validation results (`fea_modal_results.json`).

Table 13: Boundary condition effects on $n = 2$ frequency (wet shell).

Configuration	f_2 (Hz)	Shift from free (%)
Free sphere (Ritz)	4.13	0
Hemisphere simply-supported	3.91	−5.3
Hemisphere clamped	3.89	−5.8
25% surface constrained	3.66	−11.4

Key result. The maximum frequency shift from boundary conditions is 11%. This insensitivity arises because the fluid added mass dominates ($m_{\text{add}}/m_{\text{wall}} \approx 7.3$), so the system dynamics are controlled by the fluid inertia rather than the shell boundary.

3.7 Summary of Modelling Assumptions

Table 14 compiles the frequency error from each modelling simplification.

4 Code Availability

4.1 Repository

All source code, data, and scripts required to reproduce the results presented in the main paper and this supplementary material are available at:

Table 14: Impact of modelling assumptions on predicted f_2.

Assumption	Error	Dominant factor
Sphere → oblate spheroid	11–20%	Curvature distribution
Homogeneous → multilayer	<4%	Muscle tone
Inviscid → viscous fluid	<1%	Stokes boundary layer
No organs → with inclusions	<2%	Density perturbation
Linear → nonlinear	5–15%*	WBV amplitudes only
Free → constrained BC	<11%	Fluid dominates

*Only relevant for mechanical coupling ($\xi > 1\,\mathrm{mm}$).

All errors are small compared to the uncertainty in E ($S_T = 0.86$; factor-of-5 range).

https://github.com/JonathanMace/brownnote

4.2 Module Listing

Table 15 describes each module in the repository.

Table 15: Repository module listing.

Module / Script	**Description**
Analytical models (`src/analytical/`)	
`natural_frequency_v2.py`	Flexural and breathing mode frequencies for a fluid-filled sphere with added mass, pre-stress, and viscoelastic damping.
`energy_budget.py`	Self-consistent energy budget via reciprocity (absorption cross-section, radiation damping, energy-consistent displacement).
`acoustic_coupling.py`	Airborne acoustic coupling: $(ka)^n$ pressure reduction, impedance mismatch, and displacement at specified SPL.
`mechanical_coupling.py`	Whole-body vibration base-excitation model: frequency-response function and displacement at specified a_{rms}.
`oblate_spheroid_ritz.py`	Rayleigh–Ritz analysis on the full oblate spheroid with Legendre-polynomial trial functions and Gauss–Legendre quadrature.
`multilayer_wall.py`	Classical laminate theory for the multi-layer abdominal wall; effective composite properties and frequency comparison.
`viscous_correction.py`	Stokes boundary-layer analysis: viscous damping ratio, Q-factor correction, crossover viscosity, and power-law (non-Newtonian) extension.

Module / Script	Description
`organ_inclusions.py`	Hashin–Shtrikman effective medium theory for solid organ inclusions; parametric study of volume fraction and stiffness.
`nonlinear_analysis.py`	Donnell–von Kármán cubic stiffness, Duffing backbone curve, jump phenomenon, and amplitude-dependent damping.
`parametric_analysis.py`	Full factorial parametric sweep (486 configurations) over E, a, c/a, h, η.
`uncertainty_quantification.py`	Monte Carlo and Sobol sensitivity analysis (N = 10,000 samples).
`gas_pocket_resonance.py`	Minnaert bubble resonance for intestinal gas pockets.
`gas_pocket_detailed.py`	Detailed sub-resonant gas-pocket response at infrasonic frequencies.
`orifice_coupling.py`	Upper GI tract (mouth-to-stomach) acoustic coupling pathway.
`mechanotransduction.py`	PIEZO1/2 activation thresholds and SPL required for channel opening.
`dimensional_analysis.py`	Buckingham-π dimensional analysis of the coupled system.
Core library (`src/browntone/`)	
`analytical/`	Shell vibration and acoustic-mode solvers.
`fem/`	FEA mesh generation and solver wrappers.
`mesh/`	Mesh utilities for parametric geometries.
`postprocess/`	Result aggregation and figure generation.
`utils/`	Shared constants, unit conversions, I/O helpers.
`cli.py`	Command-line interface entry point.
Scripts (`scripts/`)	
`run_modal_analysis.py`	Run the full modal analysis pipeline (all modes, parametric sweep).
`run_convergence_study.py`	Mesh and quadrature convergence study for Rayleigh–Ritz solver.
`generate_jsv_figures.py`	Generate all publication figures for the JSV paper.
Data (`data/`)	
`materials/default.json`	Material property database (tissue, fluid).
`results/fea_modal_results.json`	FEA validation: analytical vs. Ritz vs. various BC configurations.
`results/uq_results.json`	Full Monte Carlo and Sobol analysis output.

4.3 Reproducing Figures and Tables

All figures and tables in the main paper can be reproduced from the repository as follows.

1. **Install the package:**

   ```
   pip install -e ".[all]"
   ```

2. **Run the full analysis pipeline:**

   ```
   python scripts/run_modal_analysis.py
   ```

   This produces all intermediate results in `data/results/`.

3. **Generate all JSV figures:**

   ```
   python scripts/generate_jsv_figures.py
   ```

   Output appears in `paper/figures/`.

4. **Run individual validation studies:**

   ```
   python -m src.analytical.oblate_spheroid_ritz
   python -m src.analytical.multilayer_wall
   python -m src.analytical.viscous_correction
   python -m src.analytical.organ_inclusions
   python -m src.analytical.nonlinear_analysis
   ```

5. **Run uncertainty quantification:**

   ```
   python -m src.analytical.uncertainty_quantification
   ```

6. **Build the paper:**

   ```
   cd paper && pdflatex main.tex && bibtex main
   pdflatex main.tex && pdflatex main.tex
   ```

7. **Run tests:**

   ```
   make test
   ```

A Docker environment is also provided for fully reproducible builds:

```
docker compose -f docker/docker-compose.yml run --rm browntone bash
```